

Fig. 3: Actual-size PCB layout of the 3W audio amplifier

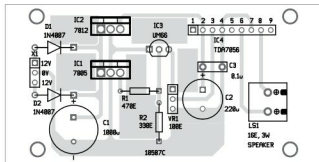


Fig. 4: Components layout for the PCB

generator UM66 and audio amplifier TDA7056. Melody generator chip UM66 (IC3) is used to provide audio input to the amplifier. It requires around 3V input. So a voltage divider network using resistors R1 and R2 is used to derive 3V from 5V output of the power supply. The output of IC3 is connected to the positive input of TDA7056 through a 100-ohm potmeter used for volume control.

Vcc pin of IC4 is connected to 12V supply along with two ca-

pacitors as shown in Fig. 2. The 3W, 16-ohm loudspeaker is connected to output pins 6 and 8 of IC4.

Construction and testing

An actual-size, single-side PCB layout for 3W audio amplifier using TDA7056 is shown in Fig. 3 and its components layout in Fig. 4.

After assembling the circuit on the PCB, enclose it in a small box. Place transformer X1 and speaker inside the box.

When you switch on the mains

power supply, UM66 chip gets activated and it generates a melody tone, which is fed to TDA7056 through potmeter VR1. TDA7056 chip amplifies the melody sound to around 3W, which can be heard from the speaker. By varying VR1, you can adjust the volume of the output tone. **EFY**



Pamarthi Kanakaraja is an assistant professor in Usha Rama College of Engineering and Technology, Andhra Pradesh



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SHOWROOM: 426, Old Lajpat Rai Market,
Deewan Hall Road,
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